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SUMMARY

B.Tech (ISE) student at IIT Kharagpur with 3 live full-stack and distributed systems projects shipped independently. Seeking an SDE internship where I can own features end-to-end, move fast, and ship production code from day one. Experienced with REST APIs, real-time systems, and cloud-integrated backends across Go, Java, and the MERN stack.

EDUCATION

B.Tech, Industrial & Systems Engineering
Indian Institute of Technology (IIT) Kharagpur

2023 – Present (*Expected May 2027*)
GPA: 8.56 / 10.0

EXPERIENCE

Software Engineering Extern
JPMorgan Chase

Apr – May 2026
Remote

- Built a Kafka-backed async transaction pipeline in Spring Boot, consumed, validated, and persisted high-volume messages to H2 via Spring Data JPA, modelling entities and balance updates across 3+ relational tables; reduced processing coupling across decoupled services.
- Developed a REST endpoint for user-balance queries via a Spring controller; connected the service to an external Incentive API using RestTemplate, incorporating responses into end-to-end transactional workflows.
- Achieved 100% test coverage across all transaction workflows with 50+ JUnit cases in Maven; pipeline processed high-volume Kafka messages with zero regression failures across 3+ integrated services.
- Collaborated in an Agile workflow, raising PRs and iterating on code review feedback; contributed across all layers from async message ingestion to DB persistence and external API integration.

PROJECTS

Synthetic-Bull — Distributed Trading Platform

Go, Gin, GORM, MySQL, WebSockets, Redis | 2025 (🔗 Live)

- Built a modular Go backend with decoupled microservices for order matching, portfolio management, and market data aggregation; applied system design principles to achieve sub-ms inter-service latency across 3+ independent services.
- Implemented a custom Limit Order Book with Price-Time Priority: $O(\log N)$ insertion via Red-Black Trees, $O(1)$ best-price retrieval, supporting partial fills across concurrent sessions.
- Engineered lock-free fan-out via goroutines and sync.WaitGroup broadcasting to 5 services; integrated a Redis low-latency caching layer for market state; exposed REST API Gateway + WebSocket endpoints for tick-by-tick candlestick rendering.
- Built a stochastic GBM market simulator generating 50–100 realistic orders/sec, driving live OHLCV candlestick aggregation with real-time frontend updates.

Collaborative Whiteboard

React.js, Firebase Firestore, Canvas API, Tailwind CSS | Aug 2025 (🔗 Live)

- Architected a multi-user drawing app supporting 10+ concurrent users with sub-100ms sync latency via Firebase Firestore real-time listeners propagating canvas state across all clients.
- Developed a stateful drawing engine with 6 tools and full undo/redo history via Canvas API and Rough.js; designed a mobile-first responsive UI with React Hooks.

MERN E-Commerce Platform

MongoDB, Express.js, React.js, Node.js, JWT, Multer | 2025 (🔗 Live)

- Implemented a full-stack e-commerce platform with JWT auth via HTTP-only cookies (XSS-safe), role-based admin controls, product image upload via Multer, and complete order + review management.
- Designed a RESTful API with Express.js and Mongoose featuring async error-handling, bcrypt hashing, and protected route guards; deployed React SPA to Netlify with a concurrent dev workflow.

TECHNICAL SKILLS

Languages: Java, Python, JavaScript (ES6+), TypeScript, Go, SQL

Frameworks & Libraries: React.js, Node.js, Express.js, Spring Boot, Gin/GORM, Tailwind CSS

Databases: MongoDB, MySQL, Firebase Firestore, Redis

Tools & Platforms: Git, Docker, GitHub Actions, Postman, AWS, GCP, Prisma, JWT

Concepts: System Design, Microservices, RESTful APIs, DSA, OOP, Agile/CI-CD

ACHIEVEMENTS & COMPETITIVE PROGRAMMING

- LeetCode:** Solved 250+ problems across arrays, trees, graphs, dynamic programming, and binary search.
- Codeforces:** Rating 1286 (Specialist), consistent performance in algorithmic contests.
- HackerRank:** 5-Star Gold Badge in Problem Solving.